Walchand College of Engineering, Sangli

(Government Aided Autonomous Institute)

7CS353 Machine Learning Lab — Assignment 3

Data Normalisation: Five Techniques and Their Visual Effects

Class, Semester	Third Year B. Tech. (CSE), Sem V — AY 2025-26
Structure	Part A: Cheat sheet (not graded) | Part B: Independent tasks (10 marks)
Duration	1 lab session (2 hrs) + take-home; submit within 1 week

0. Dataset — Generate It Yourself

Run this cell once to create and save two columns: blood pressure readings (Normal distribution) and patient waiting times (Gamma distribution). Five extreme outliers are injected into each column so you can observe how each normalisation method responds to them.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

rng = np.random.default_rng(seed=42)   # fixed — do not change
n   = 200

# Normal distribution: systolic blood pressure (mm Hg)
bp   = rng.normal(loc=120, scale=15, size=n).round(1)

# Gamma distribution: patient waiting time (minutes)
wait = rng.gamma(shape=2, scale=30, size=n).round(1)

# Inject 5 high outliers into each column
bp[rng.choice(n, 5, replace=False)]   = rng.uniform(190, 230, 5).round(1)
wait[rng.choice(n, 5, replace=False)] = rng.uniform(350, 500, 5).round(1)

df = pd.DataFrame({"bp": bp, "wait": wait})
df.to_csv("normalisation_data.csv", index=False)
print(df.describe().round(2))
df.head()
```

Data dictionary:

Column	Distribution	Real-world meaning
bp (blood pressure)	Normal (μ=120, σ=15) + 5 outliers	Systolic blood pressure (mm Hg); symmetric around healthy mean of 120
wait (waiting time)	Gamma (shape=2, scale=30) + 5 outliers	Patient waiting time (minutes); right-skewed — most patients wait a short time, a few wait very long

PART A — Quick Reference Cheat Sheet

Read before starting Part B. Not graded. Return here when you forget a formula or function.

A.1 The Five Normalisation Techniques

In each formula, x is the original column (a pandas Series or NumPy array).

Technique	Formula	Output range	Key property
Min scaling	`x - x.min()`	[0, max – min]	Shifts the distribution so the minimum is 0. Shape and spread unchanged.
Max scaling	`x / x.max()`	[min/max, 1]	Divides by the maximum. Compresses to ≤ 1 but does not shift the minimum.
Min-Max scaling	`(x - x.min()) / (x.max() - x.min())`	[0, 1]	Maps all values to [0,1]. Sensitive to outliers — a single extreme value compresses everything else.
Z-score (standard)	`(x - x.mean()) / x.std()`	mean=0, std=1	Centers at 0 with unit variance. Outliers produce large |z| values but do not disappear.
IQR / Robust	`(x - x.median()) / (x.quantile(.75) - x.quantile(.25))`	median=0	Centers at the median and scales by IQR. Robust: outliers get large values but normal data compresses less.

A.2 Python One-Liners for Each Technique

Let s = df["bp"] (a pandas Series). Each line produces a new Series — assign it to a variable.

```
s = df["bp"]                                      # or df["wait"]

bp_min  = s - s.min()                            # min scaling
bp_max  = s / s.max()                            # max scaling
bp_mm   = (s - s.min()) / (s.max() - s.min())   # min-max
bp_z    = (s - s.mean()) / s.std()               # z-score

Q1, Q3  = s.quantile(0.25), s.quantile(0.75)
bp_iqr  = (s - s.median()) / (Q3 - Q1)          # IQR / robust
```

A.3 Plotting: Histogram + Box Plot Side by Side After Each Normalisation

Use this template after every normalisation. Replace bp_mm and the title string each time.

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(11, 4))

# histogram
ax1.hist(bp_mm, bins=20, color="steelblue", edgecolor="white")
ax1.set_title("Min-Max — bp — histogram")
ax1.set_xlabel("Normalised value"); ax1.set_ylabel("Count")

# box plot
ax2.boxplot(bp_mm)
ax2.set_title("Min-Max — bp — box plot")
ax2.set_ylabel("Normalised value")

plt.tight_layout()
plt.show()
```

What to look for in the histogram: the shape of the distribution should be the same as the raw data — only the numbers on the x-axis change. What to look for in the box plot: where the median sits, how long the whiskers are, and whether outlier dots are visible.

A.4 Quick Stats After Normalisation

Function / call	What it does	Minimal example
`pd.Series.describe()`	Count, mean, std, min, quartiles, max in one call	`bp_mm_s = pd.Series(bp_mm, name="bp_mm")` `bp_mm_s.describe().round(4)`
`np.percentile(arr, [25,50,75])`	Manual quartile check	`np.percentile(bp_z, [25, 50, 75]).round(3)`
`pd.DataFrame of all versions`	Compare all normalisations in one describe()	`pd.DataFrame({"raw":s,"min_scale":bp_min, "max_scale":bp_max,"min_max":bp_mm, "z":bp_z,"iqr":bp_iqr}) .describe().round(3)`

PART B — Independent Tasks (10 marks)

Use Part A as your reference. Every task in its own cell. All plots must have a title and labelled axes.

B.1 Generate and Explore the Raw Data

Run the dataset cell from Section 0, then work only with normalisation_data.csv from here on.

B1. Load normalisation_data.csv. Print shape, head, and describe(). In a markdown cell, record for each column: the mean, median, min, max, and whether mean is above or below median. Explain what a large gap between mean and median tells you about shape.

B2. Plot a histogram and box plot of the raw bp column side by side. Repeat for the raw wait column. In a markdown cell, compare the two pairs of plots — which column looks more symmetric, and which shows a longer upper tail? Are outlier dots visible in the box plots?

B3. Compute the skewness of both raw columns using pd.Series.skew(). Record the values in a markdown cell and explain what positive skewness means visually. Which column is more skewed, and does that match what you see in the plots?

B.2 Min Scaling (x − min)

B4. Apply min scaling to both columns. Print the min and max of each result and confirm that every minimum is now exactly 0.

B5. For each column, plot a histogram and box plot side by side (label the technique and column in the title). Write one sentence per column: what changed on the x-axis compared to the raw plots, and what stayed the same?

B6. In a markdown cell, state one situation where shifting a distribution so its minimum is 0 is useful, and one limitation of this technique.

B.3 Max Scaling (x / max)

B7. Apply max scaling to both columns. Print the min and max of each result. Is the minimum of bp_max the same as the minimum of bp_mm? Why or why not?

B8. Plot histogram + box plot for each column after max scaling. In a markdown cell, compare these plots with the min scaling plots from B.2: what is different, and why does the minimum of the max-scaled column not equal 0?

B.4 Min-Max Scaling ((x − min) / (max − min))

B9. Apply min-max scaling to both columns. Confirm that every minimum is 0 and every maximum is 1.

B10. Plot histogram + box plot for each column. In a markdown cell, locate the outlier dots on the box plots: do they look more squashed towards the high end compared with the z-score and IQR plots you will draw later? (Make a note — you will return to this in B.7.)

B11. In a markdown cell, explain why min-max scaling is sensitive to outliers. What would happen to all the non-outlier data points if a single value of 10,000 were added to the wait column?

B.5 Z-Score Normalisation ((x − mean) / std)

B12. Apply z-score normalisation to both columns. Print the mean and standard deviation of each result — they should be approximately 0 and 1. Are they exactly 0 and 1? Why might there be a tiny numerical difference?

B13. Plot histogram + box plot for each column. In a markdown cell, identify the z-score value of the outliers from the box plot (read off the approximate y-position of the outlier dots). Are any beyond ±3? What does a z-score of +4 mean in plain language?

B14. Compare the z-score histogram of bp with its raw histogram from B.1. Write one sentence: how did the shape change, and how did the x-axis scale change?

B.6 IQR / Robust Scaling ((x − median) / IQR)

B15. Apply IQR scaling to both columns. Print the median and IQR of each result — the median of the normalised column should be exactly 0. Verify this.

B16. Plot histogram + box plot for each column. In a markdown cell, observe the outlier dots on the box plots: are they farther from the box compared with z-score normalisation? Explain why IQR scaling pushes outliers further out while keeping the main body of the data more compact.

B17. Compare the IQR-scaled box plot of wait with the z-score-scaled box plot of wait side by side. In a markdown cell, state which method you would prefer for a dataset with many extreme outliers and give one reason.

B.7 Summary Comparison

B18. Create a single DataFrame containing the raw column and all five normalised versions of bp. Call describe() on it and print the result. In a markdown cell, fill in this observation table: Technique | min | max | mean | median raw | ? | ? | ? | ? min scale | ... | ... | ... | ... (and so on for all five) Which two techniques produce the same shape with different x-axis scales?

B19. Draw one figure with five box plots side by side — one per normalisation technique — for the wait column. Use plt.subplots(1,5) and label each subplot with the technique name. Write two sentences: which technique most compresses the main body of the data, and which technique makes outliers look the most extreme?

B20. In a final markdown cell, write a short paragraph (4–6 sentences) answering: for a machine learning model that is sensitive to the scale of features (e.g., K-NN), which normalisation technique would you choose for a dataset that contains outliers, and why? Mention at least two techniques in your comparison.

B.8 Take-Home (Submit with Notebook)

B21. Generate a new column: bimodal = np.concatenate([rng.normal(50,5,100), rng.normal(150,5,100)]). Apply all five normalisations to it. For each, draw a histogram. Which normalisation preserves the bimodal shape most clearly? Which technique would be misleading if you assumed the output was always unimodal?

B22. Without using any pandas or NumPy normalisation helper: write your own functions min_max_scale(arr) and z_score_scale(arr) using only basic Python arithmetic. Verify that their output matches your earlier pandas results for the bp column (hint: np.allclose is useful here).

B23. Explore log transformation.

Deliverables

- One notebook RollNo_Name_Assignment4.ipynb — all Part B tasks with visible output.
- Every task in its own cell with a one-line markdown comment above it.
- Every plot must have a title and labelled axes; unlabelled plots count as incomplete.
- All written interpretations in markdown cells, in your own words.

- Viva at the start of the next session — be ready to explain any plot or markdown cell live.
- Submit as per submission instructions of assignment 1 and 2.

Assessment Rubric (10 marks — Part B only)

Performance across all Part B tasks + viva	**Marks (out of 10)**
100% — all tasks complete, plots labelled, interpretations correct, viva fluent	10
90%	9
80%	8
70%	7
60%	6
50%	5
Below 50%	Proportional (percentage ÷ 10, rounded)

A task you cannot explain in the viva does not count as completed. Plots without axis labels or titles count as incomplete. Generated code awards 0 marks.